ANSWER KEY & MARKING SCHEME SET 1

Chapter 10: **Minerals and Energy Resources (Geography)** | Class 10 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** manganese	Used in ferro-manganese
2	**(a)** light, strong and a good conductor	Combines lightness with strength
3	**(b)** limestone	It is a calcium carbonate rock
4	**(b)** transport of coal is cheaper	Bulky fuel is costly to move
5	**(d)** copper	Copper belt
6	**(a)** coal	A fossil fuel
7	**(c)** Nagercoil to Madurai	Tamil Nadu
8	**(b)** natural gas	Low carbon dioxide emission
9	**(d)** CNG	Compressed natural gas
10	**(c)** thorium for atomic energy	Thorium is a nuclear fuel

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	India is a tropical country with abundant sunshine (1); Solar power is renewable and suits remote villages (1).	1+1
12	Mining destroys forests and land (1); It pollutes air, water and soil (1).	1+1
13	Coal was formed in particular geological basins (1); It is concentrated in the Gondwana basins of eastern India (1).	1+1
14	India has large reserves of high-grade iron ore (1); Exports earn foreign exchange, especially to Japan and South Korea (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Limestone occurs in sedimentary rocks (1); It is the raw material for cement (1); It is used in smelting iron (1).	1+1+1
16	The Odisha-Jharkhand belt has Mayurbhanj and Kendujhar (1); Chhattisgarh has Bailadila (1); Karnataka has Kudremukh (1).	1+1+1
17	Peat has lowest carbon (1); Lignite and bituminous follow (1); Anthracite has the highest carbon and quality (1).	1+1+1
	OR: Geothermal energy comes from the heat of the earth (1); Tidal energy uses ocean tides in the Gulf of Khambhat (1); Both are renewable but limited to special sites (1).	1+1+1
18	India has abundant sunshine (1); Photovoltaic cells convert it to power (1); It is useful in rural and remote areas (1).	1+1+1
	OR: Mumbai High is offshore in the Arabian Sea (1); Gujarat has Ankleshwar (1); Assam has Digboi and Naharkatiya (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Conventional sources are coal, oil, gas and hydel (1); They are widely used but finite except hydel (1); Non-conventional are solar, wind, biogas, tidal (1); They are renewable and eco-friendly (1); They are costlier to set up (1).	1+1+1+1+1
	OR: Thermal power uses coal and gas (1); Hydel uses falling water (1); Nuclear uses uranium (1); Solar and wind are renewable (1); Each has limits such as pollution, displacement, risk or site (1).	1+1+1+1+1
20	Iron ore is of haematite and magnetite types (1); Odisha, Chhattisgarh and Karnataka have major deposits (1); Manganese is in Odisha (1); Bauxite is in Odisha, copper in Madhya Pradesh (1); These support steel and aluminium industries (1).	1+1+1+1+1
	OR: Oil is a liquid fuel used in transport (1); Natural gas is cleaner and used as CNG (1); Oil is mostly imported (1); Gas is found with oil (1); Both are fossil fuels and limited (1).	1+1+1+1+1

Mistakes to avoid:

- Writing that **hematite** has more iron than magnetite; **magnetite** is richer (up to 70 per cent), though hematite is used more in quantity.

- Mixing the order of coal: peat, lignite, bituminous, **anthracite** (highest carbon, best quality).
- Classifying **hydel power** as non-conventional; hydel power, coal, petroleum, firewood and natural gas are conventional.
- Calling minerals renewable; they are exhaustible and take millions of years to form.
- Confusing **veins** and **lodes** (igneous and metamorphic rocks) with **placer deposits** (alluvial sands) and sedimentary beds.
- Forgetting the sources of nuclear minerals: **uranium and thorium** occur in Jharkhand and the Aravalli ranges of Rajasthan, and thorium is also found in the **monazite sands of Kerala**.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 10: **Minerals and Energy Resources (Geography)** | Class 10 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** malleable, ductile and a good conductor of electricity	Wires and electronics
2	**(b)** solar energy	Non-conventional sources are renewable
3	**(c)** availability of water and suitable relief	Flow and head of water
4	**(b)** iron ore	Exported through Visakhapatnam
5	**(c)** contains iron	Ferrous means iron-containing
6	**(b)** narrow, unsafe tunnels	Tunnels may collapse or flood
7	**(a)** renewable and pollution-free	Sunlight is abundant in India
8	**(a)** magnetite	Best quality ore
9	**(c)** uses farm waste and gives manure	Produced from dung and waste
10	**(c)** Neyveli	Brown coal

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Limestone is used as a flux in the blast furnace (1); It removes impurities from iron ore (1).	1+1
12	Bauxite forms by decomposition of aluminium-rich rocks (1); Plateaus provide the conditions and have deposits such as Amarkantak (1).	1+1
13	Water is renewed by the water cycle (1); Falling water turns turbines without being used up (1).	1+1
14	It needs strong and steady winds (1); Only certain sites like Tamil Nadu are suitable (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Mica occurs in Jharkhand, Andhra Pradesh and Rajasthan (1); It is an insulator (1); It is used in electrical industry (1).	1+1+1
16	Biogas plants use dung and farm waste (1); They produce gas for cooking (1); Residue is good manure (1).	1+1+1
17	Bauxite is found on plateaus of Amarkantak and Maikal (1); Odisha is the leading producer (1); Aluminium is used in aircraft and utensils (1).	1+1+1
	OR: Miners crawl through narrow tunnels in Meghalaya (1); Tunnels may collapse or flood (1); Workers die, and the practice is unsafe and unregulated (1).	1+1+1
18	Manganese is used in steel making (1); It is used in ferro-manganese alloys (1); It is also used in batteries and paints (1).	1+1+1
	OR: Ferrous minerals contain iron, like iron ore and manganese (1); Non-ferrous minerals do not contain iron, like copper and bauxite (1); Ferrous minerals are three-fourths of metallic mineral production (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Minerals supply raw materials for industries (1); Iron ore feeds steel plants (1); Limestone feeds cement (1); Coal and oil give energy (1); Exports earn foreign exchange (1).	1+1+1+1+1
	OR: Thermal plants are near coalfields (1); Steel plants are near iron ore and coal (1); Aluminium plants need cheap power (1); Hydel areas attract industry (1); Transport cost decides location (1).	1+1+1+1+1
20	Minerals are finite and take long to form (1); Rich deposits are limited (1); Recycling scrap saves resources (1); Substitutes reduce pressure (1); Efficient mining techniques reduce waste (1).	1+1+1+1+1
	OR: India has abundant sunshine for solar power (1); Wind potential exists in Tamil Nadu, Gujarat (1); Biogas suits villages (1); Costs and technology limit expansion (1); Policy support can increase use (1).	1+1+1+1+1

Mistakes to avoid:

- Writing that **hematite** has more iron than magnetite; **magnetite** is richer (up to 70 per cent), though hematite is used more in quantity.

- Mixing the order of coal: peat, lignite, bituminous, **anthracite** (highest carbon, best quality).
- Classifying **hydel power** as non-conventional; hydel power, coal, petroleum, firewood and natural gas are conventional.
- Calling minerals renewable; they are exhaustible and take millions of years to form.
- Confusing **veins** and **lodes** (igneous and metamorphic rocks) with **placer deposits** (alluvial sands) and sedimentary beds.
- Forgetting the sources of nuclear minerals: **uranium and thorium** occur in Jharkhand and the Aravalli ranges of Rajasthan, and thorium is also found in the **monazite sands of Kerala**.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 10: **Minerals and Energy Resources (Geography)** | Class 10 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** they take millions of years to form and cannot be replenished quickly	Rate of formation is very slow
2	**(a)** Jharia	Gondwana coal
3	**(a)** it leads in manganese and bauxite and has rich iron ore	Rich mineral state
4	**(c)** mica	Used in electrical industries
5	**(c)** coal	Occurs in layers
6	**(a)** scrap metals	Reuse metals
7	**(a)** Ankleshwar	Onshore field
8	**(b)** of safety risks and radioactive waste	Accidents are dangerous
9	**(b)** Odisha	Panchpatmali deposits
10	**(a)** they are finite and non-renewable	Future generations need them

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	India produces little crude oil compared with its demand (1); Rising transport and industry need more oil (1).	1+1
12	Industries need electricity and fuel (1); Higher output increases demand for coal, oil and power (1).	1+1
13	It releases less carbon dioxide than coal and oil (1); It produces less smoke and pollution (1).	1+1
14	They were formed by geological processes in specific places (1); Some regions have rocks suitable for formation, others do not (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Tamil Nadu has the largest cluster from Nagercoil to Madurai (1); Andhra Pradesh, Karnataka and Gujarat also have wind farms (1); It is clean but depends on winds (1).	1+1+1
16	Natural gas is a clean fuel (1); It is used as CNG in vehicles (1); It is used in power and fertiliser plants (1).	1+1+1
17	Haematite has 60 to 70 per cent iron (1); Magnetite has over 70 per cent and is the best (1); Haematite is the most widely used (1).	1+1+1
	OR: Copper is malleable, ductile and a good conductor (1); It is used in wires and electronics (1); Balaghat in Madhya Pradesh leads (1).	1+1+1
18	Uranium and thorium are nuclear fuels (1); Monazite sands of Kerala have thorium (1); Tarapur and Kalpakkam are stations (1).	1+1+1
	OR: Thermal power uses fossil fuels and causes pollution (1); Hydel uses water and is renewable (1); Hydel plants need dams and suitable terrain (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Coal is found in Gondwana valleys of Damodar, Mahanadi and Godavari (1); It is used for power and industry (1); Oil comes from Mumbai High, Gujarat and Assam (1); Refineries make petrol and diesel (1); India imports a large amount of oil (1).	1+1+1+1+1
	OR: Energy demand is rising with growth (1); Fossil fuels are limited (1); Oil is imported (1); Conservation, renewables and efficiency are needed (1); Public transport reduces demand (1).	1+1+1+1+1
20	Minerals are metallic and non-metallic (1); Metallic ones are ferrous or non-ferrous (1); Iron ore is in Odisha, Chhattisgarh and Karnataka (1); Bauxite is in Odisha and copper in Madhya Pradesh (1); Coal and oil are energy minerals (1).	1+1+1+1+1

	OR: Mining is dangerous (1); Rat-hole mines collapse and flood (1); Land and water are polluted (1); Safety rules and inspection are needed (1); Mined land should be restored (1).	1+1+1+1+1

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